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JavaScript Drill: Part II

Lab2.1 Variables and Data Types

Objective

Declare and display variables of different data types.

Instructions

1. Declare variables: name (string), age (number), isStudent (boolean), grades (array), student (object).
2. Print them and their types using `console.log` and `typeof`.

Code:

```
let name = "Juan";
let age = 20;
let isStudent = true;
let grades = [85, 90, 92];
let student = {firstName: "Juan", lastName: "Dela Cruz"};

console.log(name, typeof name);
console.log(age, typeof age);
console.log(isStudent, typeof isStudent);
console.log(grades, typeof grades);
console.log(student, typeof student);
```

Expected Output

Juan string
20 number
true boolean [85, 90, 92]
object {firstName: "Juan", lastName: "Dela Cruz"} object

Lab 2.3: Arithmetic & Type Conversion

Objective

Perform arithmetic operations and demonstrate type conversion.

Instructions

1. Ask the user for two numbers using `prompt()`.
2. Perform addition, subtraction, multiplication, and division.
3. Display results in an alert.
4. Show conversion of a string to a number and a number to boolean.

Code:

```
let num1 = Number(prompt("Enter first number:"));
let num2 = Number(prompt("Enter second number:"));

let sum = num1 + num2;
let diff = num1 - num2;
let prod = num1 * num2;
let div = num1 / num2;

alert(`Results:
Sum = ${sum}
Difference = ${diff}
Product = ${prod}`);
```

```
Division = ${div}`);

let numString = "123";
let numValue = Number(numString);
let boolValue = Boolean(0);

console.log("String:", numString, "Type:", typeof numString);
console.log("Converted Number:", numValue, "Type:", typeof numValue);
console.log("Boolean Value of 0:", boolValue);
```

Expected Output

If the user enters 10 and 5:

Results:

Sum = 15

Difference = 5

Product = 50

Division = 2

Console:

String: 123 Type: string

Converted Number: 123 Type: number

Boolean Value of 0: false Type: boolean

Lab 2.4: Conditional Statements

Objective

Use comparison and logical operators in conditionals.

Instructions

1. Ask the user for their age.
2. If 18 or older, display “You are an adult.” Otherwise, display “You are a minor.”
3. Validate the input to ensure it’s numeric.

Code:

```
let ageInput = prompt("Enter your age:");
let ageNum = Number(ageInput);

if (!isNaN(ageNum) && ageNum >= 18) {
    alert("You are an adult.");
} else if (!isNaN(ageNum)) {
    alert("You are a minor.");
} else {
    alert("Invalid input!");
}
```

Expected Output

- If the user enters 20:
You are an adult.
- If the user enters 15:
You are a minor.
- If the user enters a non-numeric value:
Invalid input!

Lab 2.5: Loops

Objective

Practice `for` and `while` loops.

Instructions

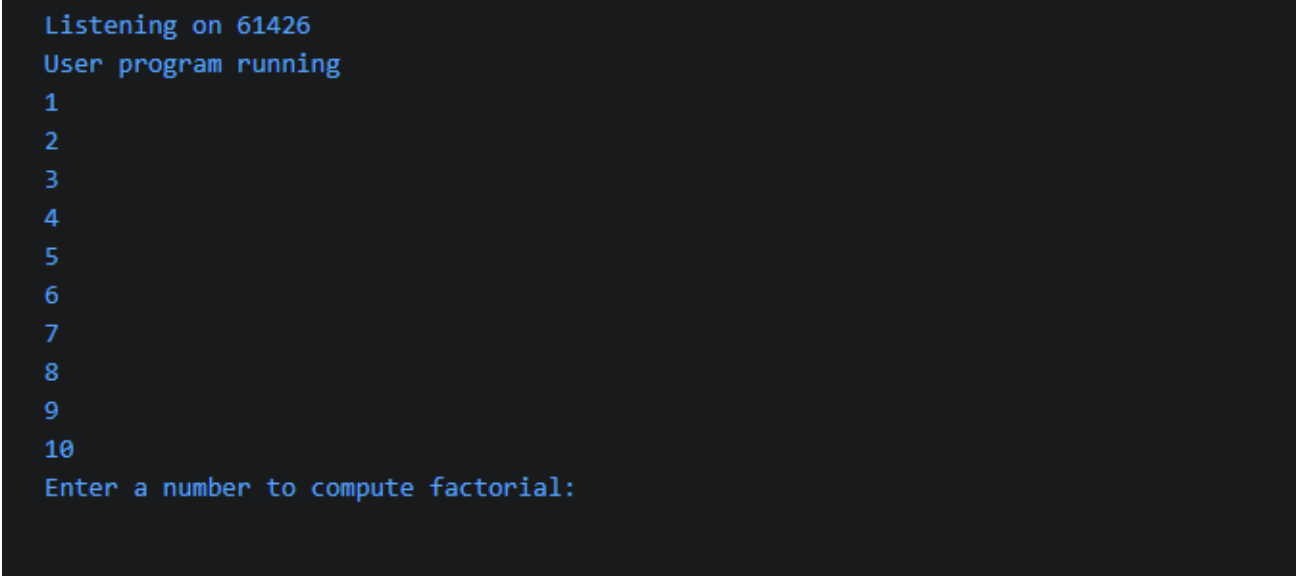
1. Use a `for` loop to display numbers 1–10 in the console.
2. Ask the user for a number and compute its factorial using a `while` loop.

Code:

```
// For loop
for (let i = 1; i <= 10; i++) {
  console.log(i);
}

// Factorial using while loop
let n = Number(prompt("Enter a number to compute factorial:"));
let fact = 1;
let i = 1;
while (i <= n) {
  fact *= i;
  i++;
}
console.log(`Factorial of ${n} = ${fact}`);
```

Expected Output



```
Listening on 61426
User program running
1
2
3
4
5
6
7
8
9
10
Enter a number to compute factorial:
```

Lab 2.6: Arrays and Objects

Objective

Manipulate arrays and objects.

Instructions

1. Create an array of fruits with three elements and add another using `.push()`.
2. Remove the first fruit using `.shift()`.
3. Create an object `student` with properties `name`, `age`, and `course`.
4. Print both the array and object to the console.

Code:

```
let fruits = ["apple", "banana", "mango"];
fruits.push("orange");
fruits.shift();
console.log(fruits);
```

```
let student = {  
  name: "Maria",  
  age: 21,  
  course: "IT"  
};  
console.log(student);
```

Expected Output

- Console shows updated fruits array.
- Console shows student object.